

7.8 Organic Synthesis (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	7. Organic Chemistry (A Level Only)
Topic	7.8 Organic Synthesis (A Level Only)
Difficulty	Easy

Time allowed: 50

Score: /38

Percentage: /100

Question 1a

Morphine is a naturally occurring opiate found in opium from poppies. Heroin is a synthetic opiate that can be synthesised from morphine. Fig. 1.1 shows the chemical structure of morphine and heroin.

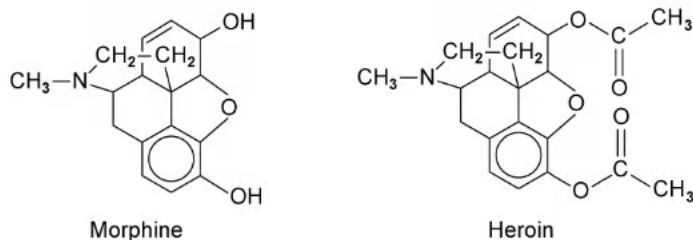


Fig. 1.1

What functional groups are found in morphine?

[2 marks]

Question 1b

Identify the functional group(s) present in heroin, but not in morphine.

[1 mark]

Question 1c

State the reagents and conditions needed to turn morphine into heroin.

[2 marks]

Question 1d

Predict and explain what you would see if morphine and heroin were separately reacted with warm acidified potassium dichromate solution.

Observation

Explanation

.....

[3 marks]

Question 2a

This question is about farnesol, a chemical found in essential oils and used in perfumes, especially in lilac perfumes. The structure of farnesol is shown in Fig. 2.1

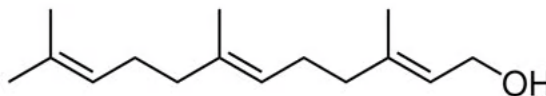


Fig. 2.1

Identify two different functional groups found in farnesol and identify their locations in Fig. 2.1.

[2 marks]

Question 2b

Describe how you would expect farnesol to react if shaken with a little bromine water.

[1 mark]

Question 2c

Farnesol is an important starting compound in organic synthesis. Fig. 2.2 shows a two-step synthesis of a compound made from farnesol.

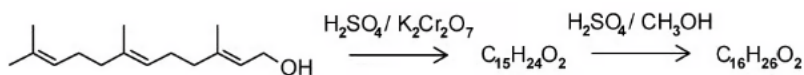


Fig. 2.2

Suggest a new functional group present in the product of the first step.

[1 mark]

Question 2d

Suggest a functional group present in the final product, but not in farnesol.

[1 mark]

Question 3a

The four step synthesis to form propan-1-ol from a ketone is outlined in Fig. 3.1.

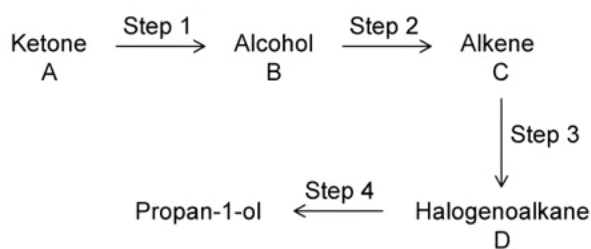


Fig. 3.1

i)
Name a possible halogenoalkane, **D**, that undergoes nucleophilic substitution in step to form propan-1-ol.

[1]

ii)
Give the reagents and conditions for step 4.

[2]

[3 marks]

Question 3b

Deduce the identity of the alkene, **C**, in Fig. 3.1 in part **(a)**.

[1 mark]

Question 3c

Suggest why the electrophilic addition reaction in step 3, in Fig. 3.1 in part **(a)**, might not be favourable in an industrial multistep reaction. You should not include economic considerations.

[1 mark]

Question 4a

Benzene and its derivatives are used in various industrial applications, including the production of plastics, synthetic fibres, rubber, dyes, and detergents.

Fig. 4.1 shows three compounds, **A**, **B** and **C**, which are all derivatives of benzene.

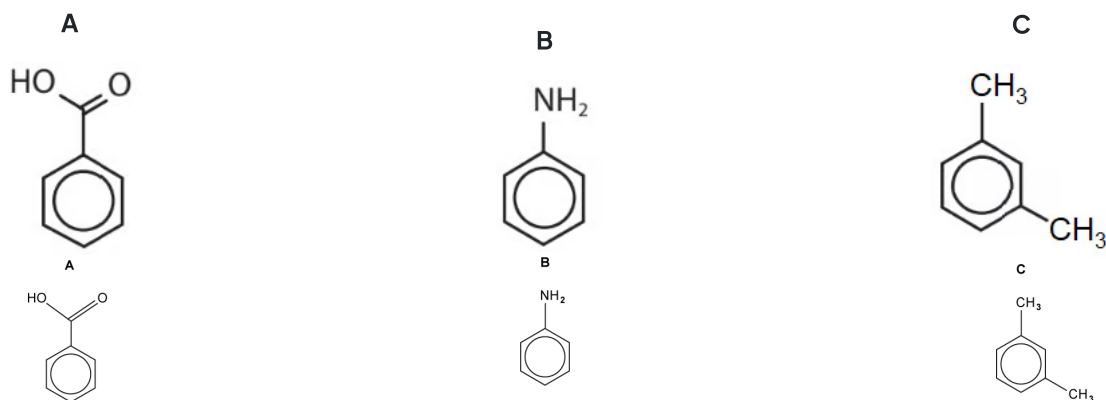


Fig. 4.1

Give the systematic names of compounds **A**, **B** and **C**.

Compound **A**

Compound **B**

Compound **C**

[3 marks]

Question 4b

Different functional groups attached to the benzene ring activate specific carbons within the ring structure and make them more likely to react.

State the positions that are activated for the following compounds in part **(a)**.

Compound **A**

Compound **B**

[2 marks]

Question 4c

Compounds **A** and **B** can be synthesised from benzene by the routes shown in Fig. 4.2.

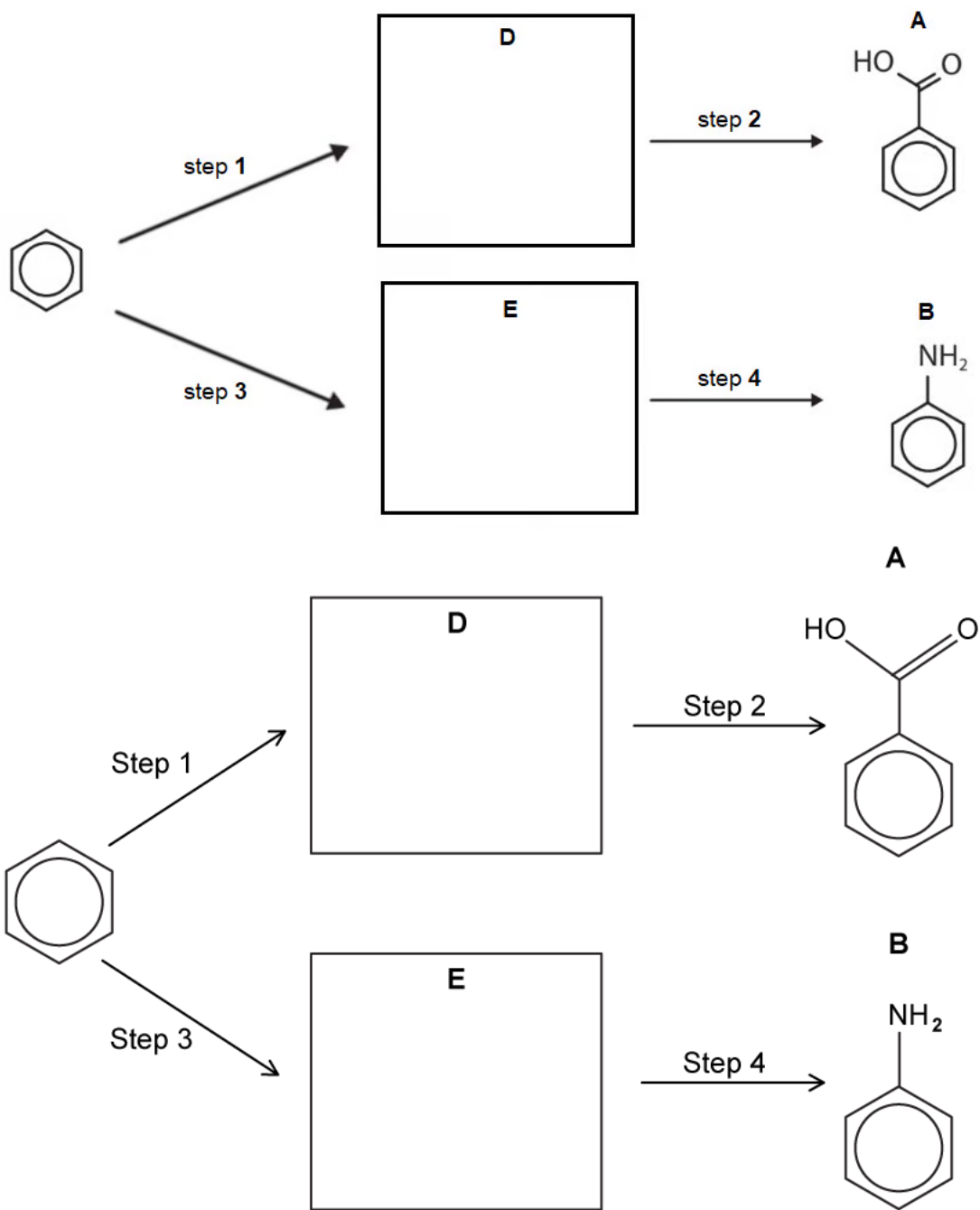


Fig. 4.2

- i)
Suggest the reagents for steps **1** - **4**.

step **1**

step 2

step 3

step 4

[4]

ii)

Draw the structures of compounds **D** and **E** in the boxes.

[2]

[4 marks]

Question 4d

The route shown in Fig. 4.3 is proposed for the synthesis of compound C.

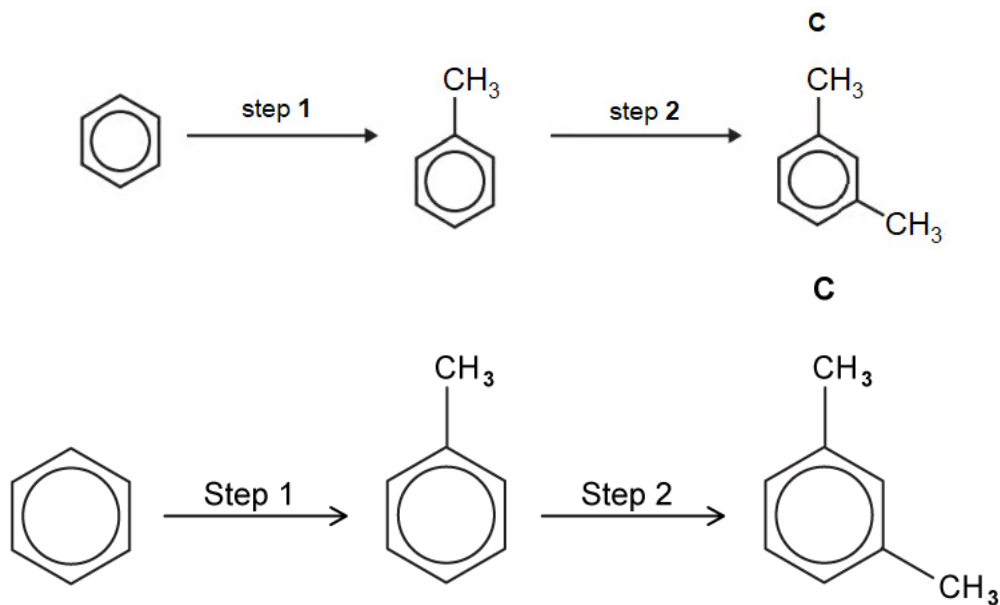


Fig. 4.3

i)

Suggest **one** set of reagents and conditions that could be used for both steps.

reagents

conditions

[2]

ii)

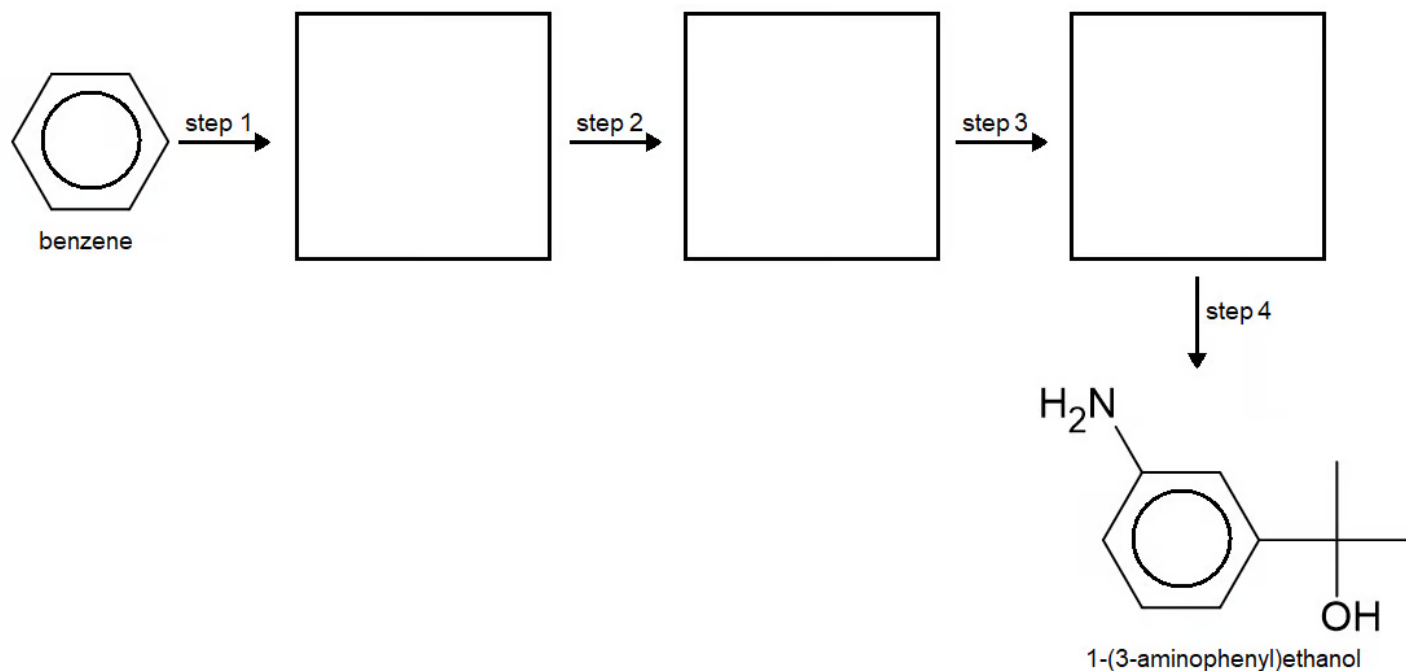
Suggest why step 2 is less likely to occur.

[1]

[3 marks]

Question 5a

1-(3-aminophenyl)ethanol can be synthesised from benzene by the route shown in Fig. 5.1.



Step 1 involves a reaction with concentrated nitric acid and concentrated sulfuric acid.

i)

Name compound **A** and state the reaction conditions for step 1.

compound **A**

reaction conditions

[2]

ii)

Name the mechanism for the conversion of benzene into compound **A** and identify the specific species that reacts with benzene.

mechanism

species

[2]

[4 marks]

Question 5b

Step 2, in Fig. 5.1 in part (a), is the reaction of compound **A** with ethanoyl chloride, CH_3COCl , in the presence of a suitable catalyst.

Identify the catalyst required for this reaction and write an equation to show how the catalyst forms the electrophile for this reaction.

[1 mark]

Question 5c

Step 3, in Fig. 5.1 in part (a), is the reaction of compound **B** with tin and hydrochloric acid, converting a nitro group into an amino group.

State **two** changes that occur to compound **B** during this reaction to show that reduction has taken place.

[1 mark]

Question 5d

Step 4, in Fig. 5.1 in part (a), is the reduction of compound **C** to 1-(3-aminophenyl)ethanol using NaBH_4 .

The structure of 1-(3-aminophenyl)ethanol is repeated in Fig. 5.2.

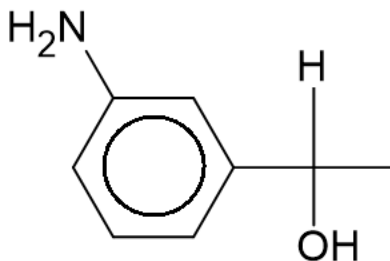


Fig. 5.2

i)

Draw a circle around the aliphatic functional group that is formed **after** the reaction of compound **C** with NaBH_4 .

[1]

ii)

Draw the structure of compound **C**.

[1]

[2 marks]

Model Answers: Easy

1a

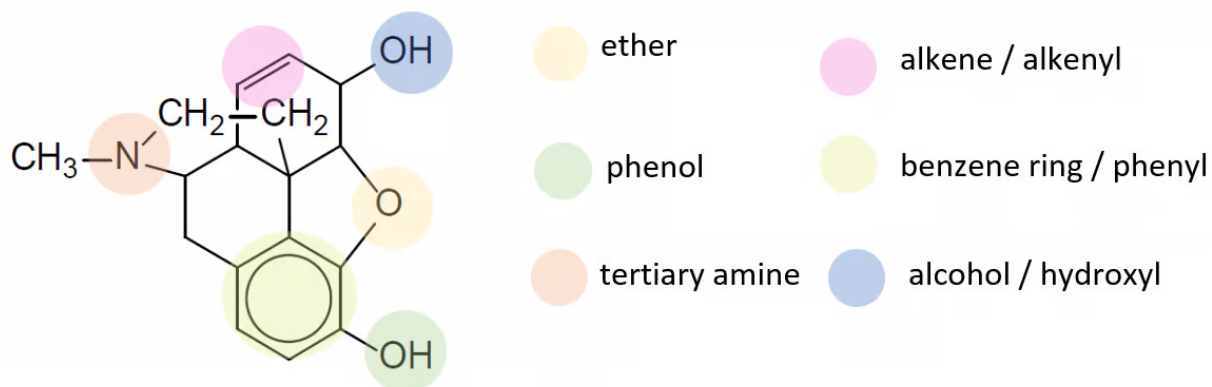
a) The functional groups found in morphine are:

TWO from: [2 marks]

- alcohol / hydroxyl;
- alkene / alkenyl;
- phenol;
- (tertiary) amine;
- ether;

[Total: 2 marks]

- You are not expected to know the ether functional group, but you may have come across it so it would be an acceptable answer
- The location of the functional groups are:



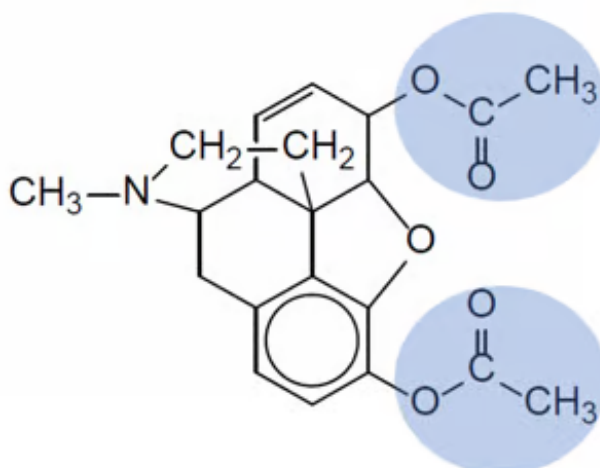
1b

b) The functional group(s) present in heroin, but not in morphine:

- Ester / ethanoate; [1 mark]

[Total: 1 mark]

- An ester group is RCOOR , in this case, ethanoate, CH_3COO^- (rest of molecule)
- Here they are on the molecule:



1c

c) The reagents and conditions needed to turn morphine into heroin are:

Reagent:

- Ethanoic acid / CH_3COOH / Ethanoic Anhydride / $(\text{CH}_3\text{CO})_2\text{O}$; [1 mark]

Conditions:

- Concentrated sulfuric acid / H_2SO_4 / concentrated hydrochloric acid / HCl / heating; [1 mark]

[Total: 2 marks]

- You may not have come across the synthesis of heroin, but you would be expected to use your knowledge and understanding of synthetic routes to deduce the reagents and conditions needed to convert one functional group into another
- You should recognise that this reaction is an esterification
- The inorganic acid is a catalyst in this reaction and it does not matter whether you use sulfuric or hydrochloric as long as it is concentrated
- Reaction conditions can include solvents, catalysts or particular temperatures and pressures

1d

d) How morphine and heroin would react with warm acidified potassium dichromate solution:

- Morphine would turn the potassium dichromate solution from orange to green; [1 mark]
- Heroin would not react / no change / stay orange with potassium dichromate; [1 mark]
- Morphine contains a (secondary) alcohol group; [1 mark]

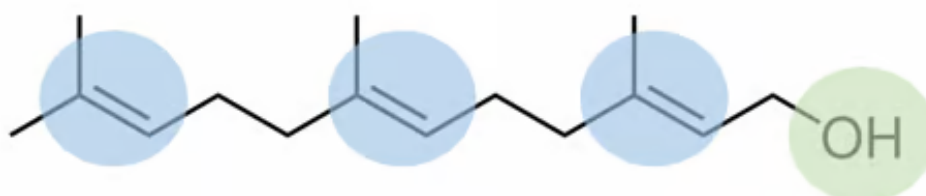
[Total: 3 marks]

- Primary and secondary alcohols will react with potassium dichromate solution as they can be oxidised
- Don't forget to give the before and after colours when describing test results

2a

a) Two functional groups found in farnesol are:

- alkene/alkenyl (highlighted in blue); [1 mark]
- (primary) alcohol/hydroxyl (highlighted in green); [1 mark]



[Total: 2 marks]

- You should be able to identify individual functional groups in structural and skeletal formulae diagrams
- Strictly speaking, the functional groups are called alkenyl and hydroxyl, respectively, and the families are called alkene and alcohol, but you wouldn't be marked down in an exam for this

2b

b) How you would expect farnesol to react with bromine water:

- The bromine water would decolourise; [1 mark]

[Total: 1 mark]

- This is a test for unsaturated compounds containing the alkene functional group
- All alkenes will decolourise bromine water, though its important to use only a few drops of bromine water as the colour may fade by not disappear completely if you use too much
- Sometimes the bromine is dissolved in cyclohexane rather than water as it can be difficult for insoluble organic compounds to react sufficiently with the aqueous bromine water, but the result is the same

2c

- c) A functional group present in the product of the first step is:
- Carboxylic acid; [1 mark]

[Total: 1 mark]

- The reagent in the first step is acidified potassium dichromate solution which is an oxidising agent
- Primary alcohols react with this oxidising agent to form aldehydes and carboxylic acids
- The intermediate has one more oxygen and two fewer hydrogens than farnesol, which is consistent with it being a carboxylic acid:
 - e.g. $\text{CH}_3\text{CH}_2\text{OH} + [\text{O}] \rightarrow \text{CH}_3\text{COOH} + \text{H}_2\text{O}$

2d

- d) A functional group present in the final product, but not in farnesol, is:
- Ester; [1 mark]

[Total: 1 mark]

- The reagent is a mixture of sulfuric acid and methanol which can be used in esterification reactions
- The product contains one more carbon and two more hydrogens, which is consistent with a (methyl) ester being formed
 - e.g. $\text{CH}_3\text{COOH} + \text{CH}_3\text{OH} \rightarrow \text{CH}_3\text{COOCH}_3 + \text{H}_2\text{O}$

3a

a)

i) The halogenoalkane, **D**, that undergoes nucleophilic substitution in step 4 to form propan-1-ol is:

- 1-chloropropane
- OR**
- 1-bromopropane; [1 mark]

ii) The reagents and conditions for step 4 are:

- Aqueous sodium hydroxide / NaOH (aq)
- OR**
- Aqueous potassium hydroxide / KOH (aq); [1 mark]
- Heat under reflux; [1 mark]

[Total: 3 marks]

- You are told that the reaction to form propan-1-ol is nucleophilic substitution
 - This means that in order to form propan-1-ol, an OH group **must** have substituted a halogen on carbon-1
 - This also suggests that the reagent is sodium hydroxide
- For nucleophilic substitution to occur, the reaction must be completed with aqueous sodium hydroxide
 - Otherwise, elimination would have occurred if alcoholic / ethanolic sodium hydroxide was used
- The conditions are a fact that you simply need to know - heat under reflux

3b

b) The identity of the alkene, **C**, in Fig. 3.1 is:

- Propene; [1 mark]

[Total: 1 mark]

- This can be deduced from a correct answer to part (a) (i) or from the final product name, propan-1-ol
- Both of these suggest that the carbon chain involved has 3 carbons
- There is only one alkene with 3 carbons, propene

3c

c) The electrophilic addition reaction in step 3 might not be favourable in an industrial multistep reaction because:

- The reaction produces more than one product / a major and minor product

OR

The products will need to be purified

OR

The reaction has low atom economy (as a second / unwanted product is formed); [1 mark]

[Total: 1 mark]

- This question tests that you know
 - Reduce the amount of steps (not applicable to this question)
 - Have a high atom economy wherever possible, which links to producing as little waste / byproducts as possible
 - Electrophilic addition to unsymmetrical alkenes can form major and minor products

4a

- This can be deduced from a correct answer to part (a) (i) or from the final product name, propan-1-ol
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 - Reduce the amount of steps (not applicable to this question)
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4a

a) The systematic names of compounds **A**, **B** and **C** are:

- Compound **A** = benzoic acid; [1 mark]
 - Compound **B** = phenylamine
- OR**
- Aminobenzene; [1 mark]
- Compound **C** = 1,3-dimethylbenzene; [1 mark]

[Total: 3 marks]

- Compound **A**
 - The functional groups of this molecule are the benzene ring and the

carboxylic acid

- The carboxylic acid always provides an -oic acid suffix to the name
- Therefore, compound **A** is benzoic acid
- Compound **B**
 - The functional groups of this molecule are the benzene ring and the amine / amino group
 - The amine / amino group can provide:
 - An amino- prefix
 - This means that one possible name for compound **B** is aminobenzene
 - Or an -amine suffix to the name
 - This means that the benzene ring should be called phenyl
 - Therefore, another possible name for compound **B** is phenylamine
 - Compound **B** is often called by its common name aniline but this will not gain the mark because it is not a systematic name
- Compound **C**
 - The functional groups of this molecule are the benzene ring and the two methyl groups
 - The two methyl groups mean that the name is prefixed with dimethyl-
 - **Remember:** You must give numbers to show the positions of the methyl groups
 - If the top methyl group is carbon-1, then the second methyl group is on carbon-3 because the functional groups are numbered to give the lowest possible numbers
 - Therefore, compound **C** is 1,3-dimethylbenzene

4b

b) The positions that are activated for the following compounds in part **(a)** are:

- Compound **A** = 3 / three
AND
5 / five; [1 mark]
- Compound **B** = 2 / two
AND
4 / four
AND
6 / six; [1 mark]

[Total: 2 marks]

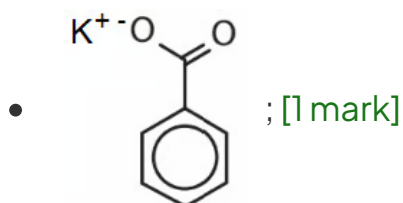
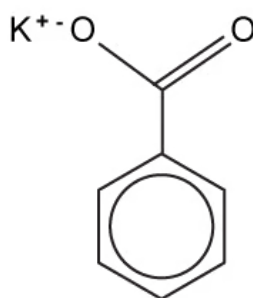
- You need to know the positions that are activated by electron-withdrawing groups such as COOH and electron-donating groups such as NH₂

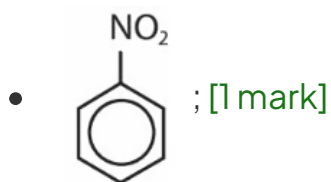
4c

c)

i) The reagents for steps **1** – **4** are:

- Step **1** = (hot alkaline) potassium manganate(VI) / KMnO₄; [1 mark]
- Step **2** = (dilute) hydrochloric acid / HCl; [1 mark]
- Step **3** = (concentrated) sulfuric acid
AND
(Concentrated) nitric acid; [1 mark]
- Step **4** = tin / Sn
AND
Hydrochloric acid / HCl; [1 mark]

ii) The structures of compounds **D** and **E** are:Compound **D**:Compound **D**Compound **E**:



Compound **E**



[Total: 6 marks]

- The conversion of benzene to benzoic acid / compound **A** is an oxidation reaction
 - Benzene is heated under reflux with hot alkaline potassium manganate(VI) to form compound potassium benzoate / compound **D**
 - Treating potassium benzoate / compound **D** with hydrochloric acid then forms benzoic acid / compound **A**
- The conversion of benzene to aminobenzene / phenylamine / compound **B** is a nitration reaction followed by reduction
 - Concentrated sulfuric acid and concentrated nitric acid are used to form the nitronium NO_2^+ ion
 - Benzene undergoes electrophilic substitution with the nitronium NO_2^+ ion to form nitrobenzene / compound **E**
 - Treating nitrobenzene / compound **E** with tin / Sn and hydrochloric acid / HCl then reduces the nitrobenzene / compound **E** to form aminobenzene / phenylamine / compound **B**
- You can also be expected to:
 - Give any necessary conditions, e.g. step **1** is heat / reflux
 - Name compounds **D** and **E**
 - State the types of reactions occurring at different steps, e.g. oxidation, reduction, nitration, electrophilic substitution
 - State / draw the mechanism for appropriate reaction steps

4d

d)

i) **One** set of reagents and conditions that could be used for both steps is:

- Reagents = Chloromethane / CH_3Cl ; [1 mark]
- Conditions = Aluminium trichloride / AlCl_3 catalyst; [1 mark]

ii) Step **2** is unlikely to occur because:

- The methyl group (on methylbenzene) activates positions 2, 4 and 6

OR

The methyl group (on methylbenzene) does not activate positions 3; [1 mark]

[Total: 3 marks]

- Both steps that are proposed in Fig. 4.3 are Friedel-Crafts alkylation reactions
 - These require an appropriate chloroalkane in the presence of an AlCl_3 catalyst
 - The AlCl_3 catalyst helps form the CH_3^+ electrophile by removing the chlorine from the chloroalkane
- Step **1** is likely to happen as the benzene has no attached functional groups that may affect the reaction
- Step **2** is less likely to occur because there is now a methyl group attached to the benzene ring
 - Methyl groups are electron-donating which means that they activate positions 2, 4 and 6 on the ring
 - Therefore, the formation of 1,3-dimethylbenzene / compound **C** is unlikely

5a

a)

i) The name of compound **A** and the reaction conditions for step 1 are:

- Compound **A** = nitrobenzene; [1 mark]
- Reaction conditions = reflux / heat

AND

Any temperature between 25 and 60 °C; [1 mark]

ii) The mechanism for the conversion of benzene into compound **A** and the reactive species are:

- Mechanism = electrophilic substitution; [1 mark]
- Species = NO_2^+ / nitronium ion; [1 mark]

[Total: 4 marks]

- The reaction of benzene with concentrated nitric acid and concentrated sulfuric acid is nitration
 - This is completed under reflux at 25 – 60 °C
 - You need to know the reagents and conditions for many organic reactions, so take the time to learn them
- Concentrated nitric acid and concentrated sulfuric acid react together to form the nitronium ion
 - The nitronium ion is an electrophile
- The nitronium ion replaces a hydrogen on the benzene ring
 - Therefore, this reaction is electrophilic substitution
- You could be asked to draw the mechanism for this reaction

5b

b) The catalyst required for this reaction and the equation for the formation of the electrophile are:

- Aluminium trichloride / AlCl_3

5a

a)

i) The name of compound **A** and the reaction conditions for step 1 are:

- Compound **A** = nitrobenzene; [1 mark]
- Reaction conditions = reflux / heat

AND

Any temperature between 25 and 60 °C; [1 mark]

ii) The mechanism for the conversion of benzene into compound **A** and the reactive species are:

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[Total: 4 marks]

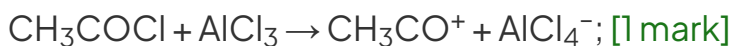
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- The nitronium ion replaces a hydrogen on the benzene ring
 - Therefore, this reaction is electrophilic substitution
- You could be asked to draw the mechanism for this reaction

5b

b) The catalyst required for this reaction and the equation for the formation of the electrophile are:

- Aluminium trichloride / AlCl_3

AND



[Total: 1 mark]

- This is another example of an electrophilic substitution reaction, that you could be asked to draw a mechanism for
- You need to know the reagents and conditions for Friedel-Crafts alkylation and acylation
 - Both reactions require an appropriate halogenoalkane and a halogen carrier catalyst such as AlCl_3
 - **Careful:** The catalyst that you choose must contain the same halogen as the reacting molecule
- You should also be able to explain how the catalyst forms the electrophile

5c

c) **Two** changes that occur to compound **B** during this reaction to show that reduction has taken place are:

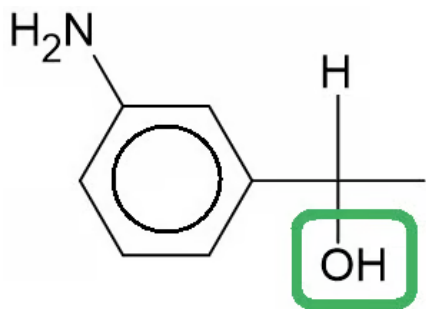
- Oxygen is lost
AND
Hydrogen is gained; [1 mark]

[Total: 1 mark]

- You do not need to know the structure of compound **B** to answer this question
 - The key piece of information is that a nitro group / NO_2 is being converted into an amine group, NH_2
 - Therefore, oxygen is being lost and hydrogen is being gained
- **Remember:** There is more than one way to describe reduction:
 - Loss of oxygen
 - Gain of hydrogen
 - Loss of electrons

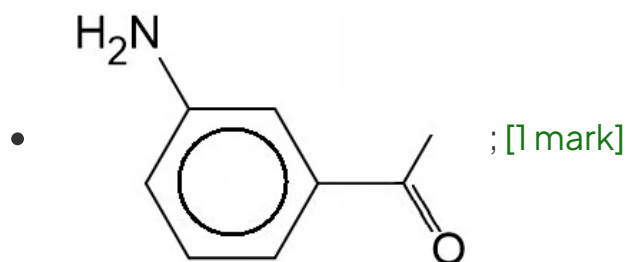
5d

d) The aliphatic functional group that is formed **after** the reaction of compound **C** with NaBH_4 is:



- OH functional group correctly identified; [1 mark]

ii) The structure of compound **C** is:



[Total: 2 marks]

- NaBH_4 is a reducing agent that is used to reduce:
 - Aldehydes $\rightarrow$ primary alcohols
 - Ketones $\rightarrow$ secondary alcohols